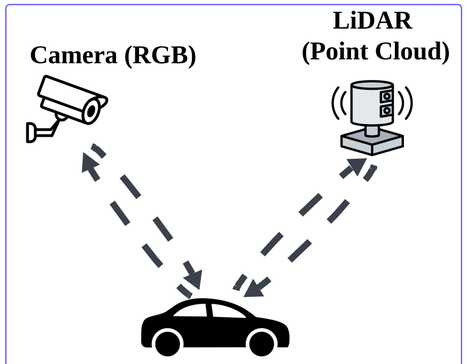
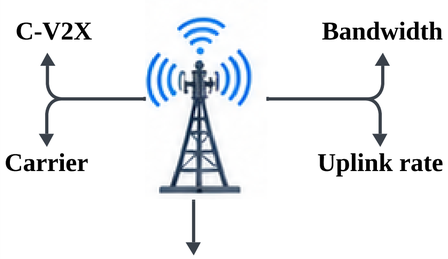
## 1. Inputs & Operating Conditions

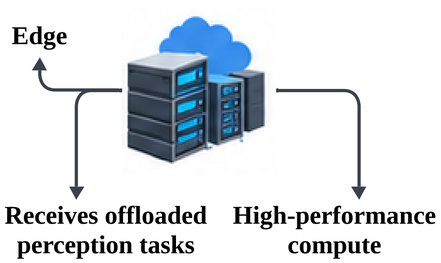
### Autonomous Vehicle & Sensors




### C-V2X Communication (V2I)



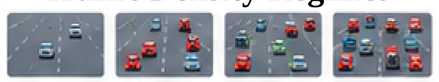

### Edge / MEC Server

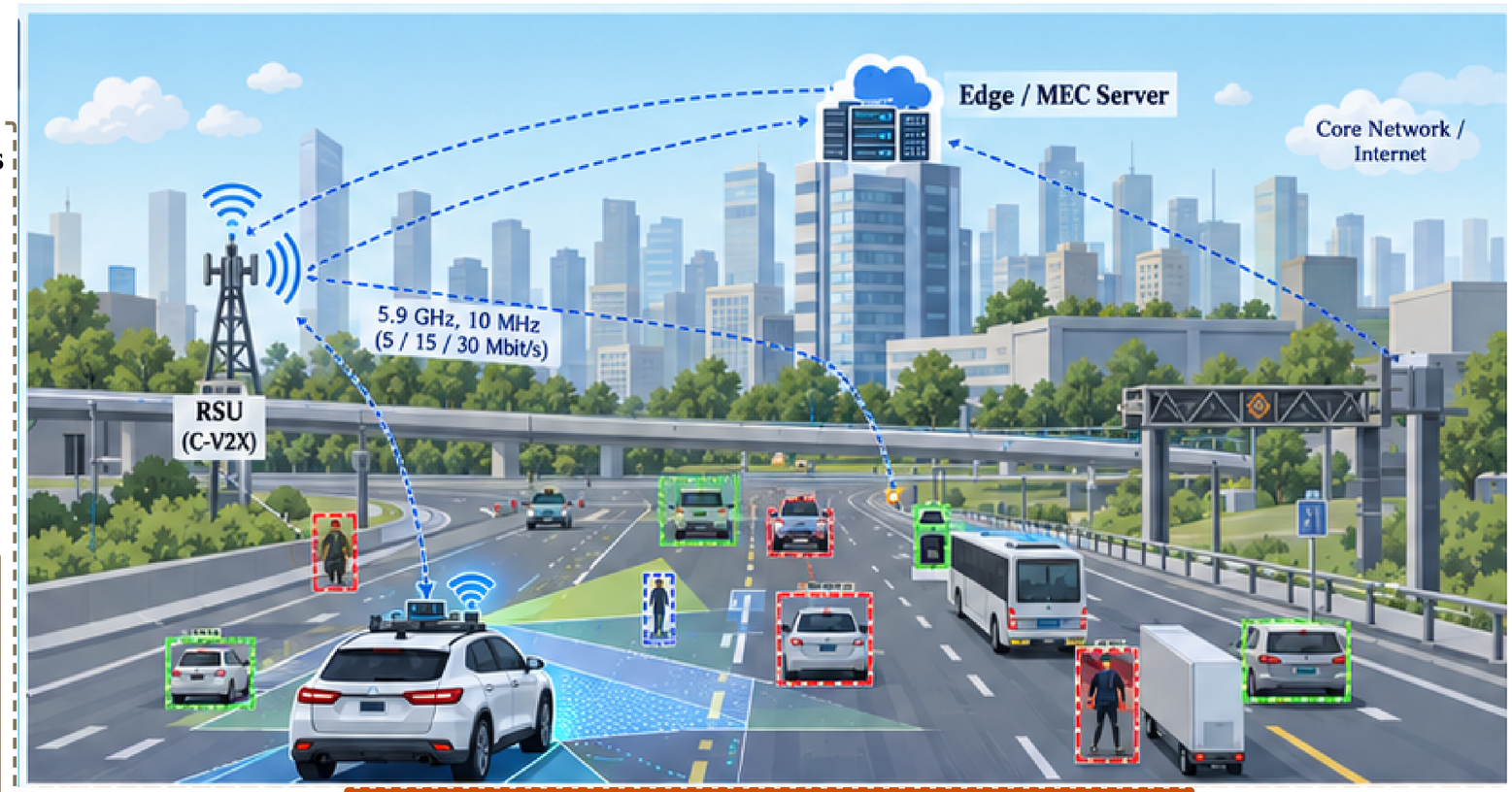



### Environment Conditions




### Traffic Density Regimes



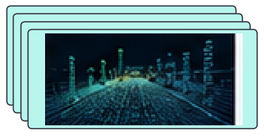

## 2. Perception-Aware Offloading and Optimization Pipeline

### 2.1 Sensor Input

Camera image



LiDAR point cloud

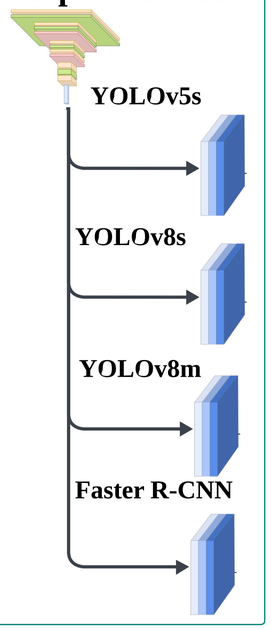


### 6 Gated Online XAI Feedback

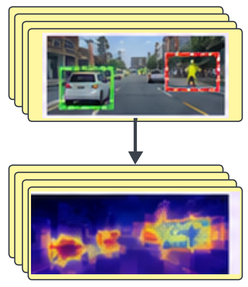
Camera image



### 2.2 Candidate Perception Models




### 2.3 Predictive Uncertainty Estimation




### 2.6 Gated Online XAI Feedback



### 2.4 Context-Aware Offloading Decision

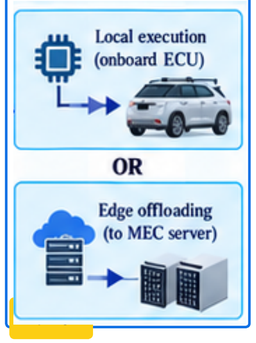
Uncertainty level

Link quality

Server load

Task complexity

### 2.5 Execution Decision




### 2.7 Adaptive Control Policy

- Minimize latency and energy
- Preserve accuracy (mAP)
- Satisfy 100{ms} deadline
- Optimal model & offloading action

## 3. Key Results / Outcomes

| | |
|---|---|
| Mean latency | 49.3 ms |
| Energy (30,000 tasks) | 38.9 kJ |
| Energy per task | 1.30 J |
| mAP@0.5 | 83.1% |
| Deadline violations | 0.6% |
| Offloaded share | 78.1% |
| Task success | 99.5% |
| Link delivery success | 99.7% |
| vs energy-aware DRL (lower latency) | 16.0% |
| vs energy-aware DRL (lower energy) | 14.3% |

## 4. Impact / Applications

- Green intelligent transportation
- Energy-efficient connected autonomous vehicles
- Reliable edge-assisted perception
- Real-time smart-road operation